Exercise 2 Report: Recovery Region for 3DOF Helicopter

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1 Plant Model and Constraints

Considering the linearized 3DOF helicopter model:

$$\dot{x}(t) = Ax(t) + Bu(t), \tag{1}$$

with the following constraints:

$$|x_1| \leq 0.5233,\ |x_2| \leq 0.174,\ |x_3| \leq 3.14,\ |x_5| \leq 0.174,\ |u_1| \leq 24,\ |u_2| \leq 24. \tag{2}$$

and initial condition:

$$x(0) = X_0 = \begin{bmatrix} 0.479 & 0.0872 & -0.8726 & 0 & 0 & 0 \end{bmatrix}^T. \tag{3}$$

The objective is to design a feedback gain K that respects these constraints and keeps the system inside the safety region.

2 Exercise 2.1 :LQR Design

The LQR law is

$$u(t) = -Kx(t), \tag{4}$$

with cost function

$$J = \int_0^\infty \left(x^T Q x + u^T R u\right) dt. \tag{5}$$

Two cases were considered, with results shown below:

- Case 1: $Q = \text{diag}(100, 1, 10, 0, 0, 2)$ and $R = \text{diag}(10, 10)$.

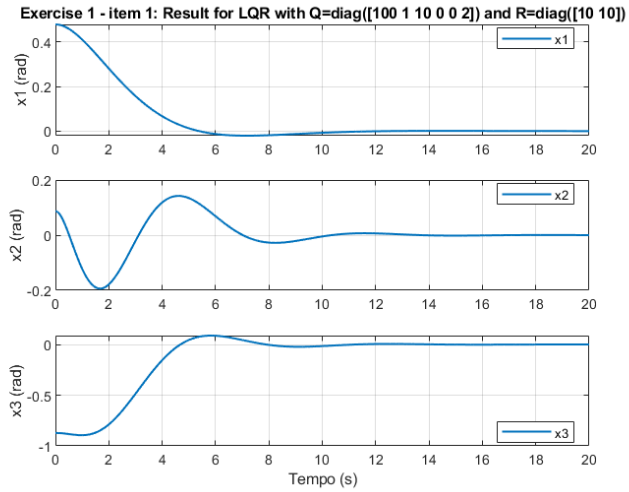


Figure 1: Result for LQR with R = 10

- Case 2: $Q = \text{diag}(100, 1, 10, 0, 0, 2)$ and $R = \text{diag}(0.25, 0.25)$.

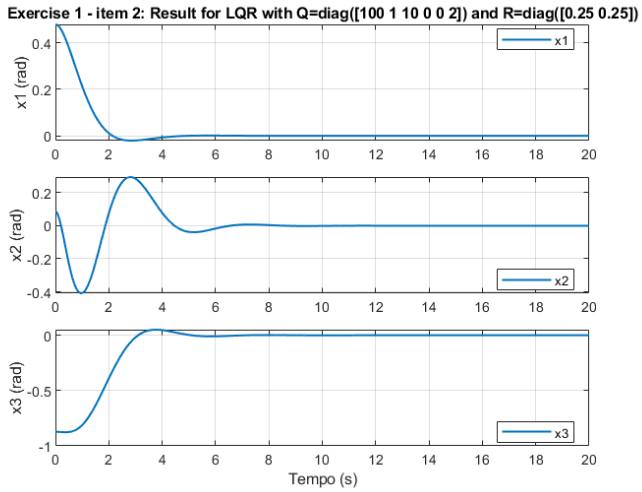


Figure 2: Result for LQR with R = 0.25

2.1 2.1: Results and Analysis

Table 1: Performance comparison for both LQR cases (Exercise 2.1).

Metric	Case 1 ($R = 10I$)	Case 2 ($R = 0.25I$)
Settling time of x_1 (s)	4.740	1.890
Overshoot of x_1 (rad)	0.02069	0.02070
Maximum deviation of x_1 (rad)	0.47901	0.47900
Settling time of x_2 (s)	12.570	7.960
Overshoot of x_2 (rad)	0.19390	0.40888
Maximum deviation of x_2 (rad)	0.19390	0.40888
Settling time of x_3 (s)	7.010	4.100
Overshoot of x_3 (rad)	0.08777	0.05057
Maximum deviation of x_3 (rad)	0.89476	0.87620
Settling time of quadratic error (s)	3.730	2.140

Interpretation of Exercise 2.1 results:

- Case 2 shows faster convergence in x_1, x_2, x_3, and in the quadratic error.
- For x_2, both cases present high peak values, with a larger peak in Case 2.
- Case 1 is more conservative overall, while Case 2 prioritizes response speed. This shows that whenever we have lower R, the faster will be the settling time because we do not penalize high inputs u

3 Exercise 2: Recovery Region Analysis

For both controllers designed in Exercise 2.1 (Case 1 and Case 2), the next step is to evaluate their recovery regions under the same state and input constraints.

3.1 Recovery Region Definition

Let the admissible set be

$$\mathcal{X}_{\text{safe}} = \{x : \ |x_1| \leq 0.5233, \ |x_2| \leq 0.174, \ |x_3| \leq 3.14, \ |x_5| \leq 0.174\} \quad (6)$$

and

$$\mathcal{U}_{\text{safe}} = \{u : \ |u_1| \leq 24, \ |u_2| \leq 24\}. \quad (7)$$

For each feedback law $u = -K_i x$ ($i \in \{1, 2\}$), the recovery region is the set of initial states whose trajectories remain admissible and converge to the equilibrium.

3.2 How Exercise 2.2 Was Computed

First, for each controller gain K, the closed-loop matrix is defined as

$$A_{\text{cl}} = A - BK. \tag{8}$$

Then, all state and input constraints are converted into linear forms

$$\alpha_j^T x \leq 1, \tag{9}$$

where the vectors α_j are created exactly as in the code:

- α_1 to α_8 represent the state bounds on x_1, x_2, x_3, and x_5.
- α_9 to α_{12} represent input bounds using $u = -Kx$, written as transformed state inequalities through $-K^T[\pm 1/24,\ 0]^T$ and $-K^T[0,\ \pm 1/24]^T$.

The optimization problem to be solved in the code is

$$\begin{aligned} \min_{Q} \quad & -\log\det(Q) && (10)\\ exts.t. \quad & QA_{\text{cl}}^T + A_{\text{cl}}Q \preceq 0, && (11)\\ & Q \succeq \varepsilon I, && (12)\\ & \alpha_j^T Q \alpha_j \leq 1, \quad j = 1, \ldots, 12. && (13) \end{aligned}$$

After solving (SDPT3 solver), the matrix returned by the script is

$$P = Q^{-1}, \tag{14}$$

and the certified recovery region is

$$\mathcal{E} = \{x : x^T P x \leq 1\}. \tag{15}$$

Finally, the projections shown in the figures are obtained by restricting to each required plane (x_1–x_4, x_2–x_5, x_3–x_6) while setting the other states to zero.

3.3 Recovery Region Projections

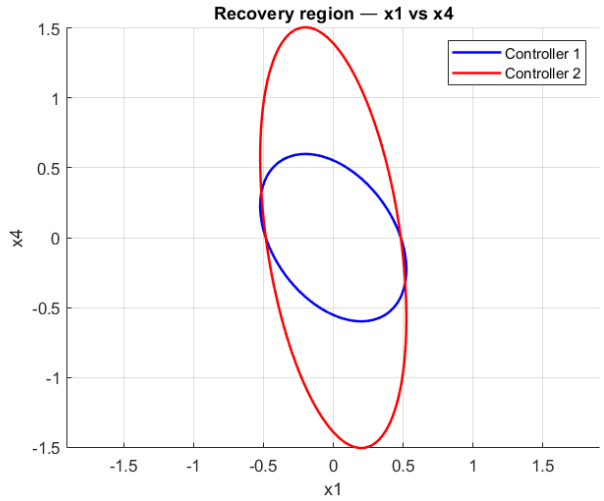


Figure 3: Recovery region projection on the x_1–x_4 plane (Case 1 and Case 2 on the same plot).

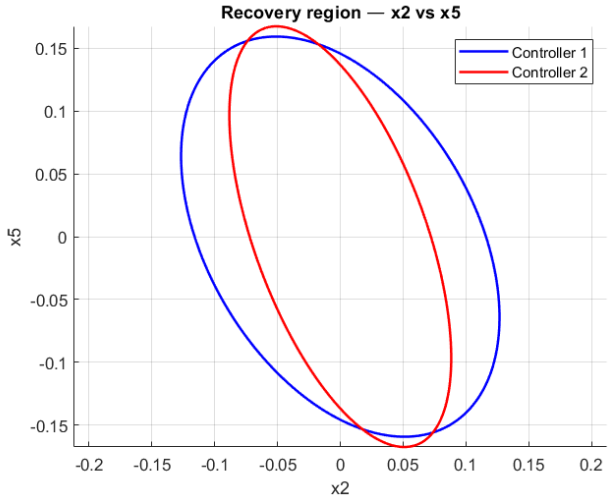


Figure 4: Recovery region projection on the x_2–x_5 plane (Case 1 and Case 2 on the same plot).

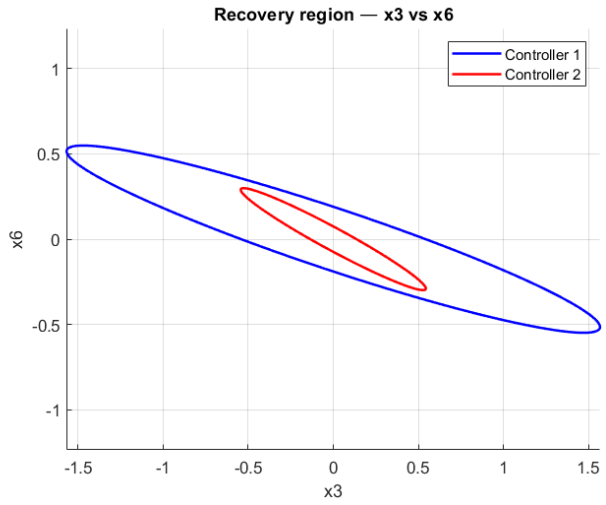


Figure 5: Recovery region projection on the x_3–x_6 plane (Case 1 and Case 2 on the same plot).

3.4 Results for Exercise 2.2

Using the computed projections in the plane order [1 4], [2 5], [3 6], the areas are:

Table 2: Projected recovery-region areas from Exercise 2.2.

Projection plane	Controller 1 (area1)	Controller 2 (area2)
x_1–x_4	0.9097	2.2879
x_2–x_5	0.0580	0.0379
x_3–x_6	0.9336	0.1272

From these values, Controller 2 provides a larger region on x_1–x_4, while Controller 1 provides larger regions on x_2–x_5 and x_3–x_6.

4 Exercise 3: Maximal-Volume Recovery Region

In Exercise 3, we compute the maximal-volume recovery region for each controller designed in Exercise 2.1, then plot the corresponding projections on the required state planes.

4.1 Optimization Objective

For each closed-loop system $\dot{x} = (A - BK_i)x$, the recovery region is approximated by an invariant ellipsoid

$$\mathcal{E}_i = \{x : x^T P_i x \leq 1\}, \quad P_i \succ 0, \tag{16}$$

that satisfies all state and input constraints. The maximal-volume ellipsoid is obtained by solving an optimization problem equivalent to maximizing $\det(P_i^{-1})$.

4.2 How the maximal volume was computed

The decision variables are Q and Z and the objective is to get

$$\min -\log\det(Q_{\max}), \tag{17}$$

which is equivalent to maximizing the ellipsoid volume.

The stability constraint is imposed as

$$Q_{\max}A^T + AQ_{\max} + Z_{\max}^T B^T + BZ_{\max} \preceq -\epsilon I, \tag{18}$$

State constraints are encoded as

$$\alpha_k^T Q_{\max} \alpha_k \leq 1, \quad k = 1, \ldots, 8, \tag{19}$$

using the vectors α_k generated from bounds on x_1, x_2, x_3, and x_5.

Input constraints are handled by $b_{u,j}$ $(j = 1, \ldots, 4)$:

$$\begin{bmatrix} 1 & b_{u,j}^T Z_{\max} \\ Z_{\max}^T b_{u,j} & Q_{\max} \end{bmatrix} \succeq 0. \tag{20}$$

These constraints enforce $|u_1| \leq 24$ and $|u_2| \leq 24$ over the certified region.

After solving with SDPT3, the controller is recovered as

$$P = Q_{\max}^{-1}, \qquad K_{\max_vol} = -Z_{\max} P. \tag{21}$$

Therefore, the maximal-volume recovery region is

$$\mathcal{E}_{\max} = \{x : x^T P x \leq 1\}. \tag{22}$$

For plotting, each required plane uses index pairs $(1, 4)$, $(2, 5)$, and $(3, 6)$. The corresponding 2D ellipse is extracted with $P_{ii} = P(\text{idx}, \text{idx})$, then drawn from $z^T P_{ii} z \leq 1$

4.3 Results for Exercise 3

Using the script output for Exercise 2.3, the projected recovery-region areas were obtained as:

Table 3: Projected recovery-region areas from Exercise 2.3.

Projection plane	Area
x_1–x_4	2.9945
x_2–x_5	0.0749
x_3–x_6	6.0726

4.4 Plane Projections of Maximal-Volume Regions

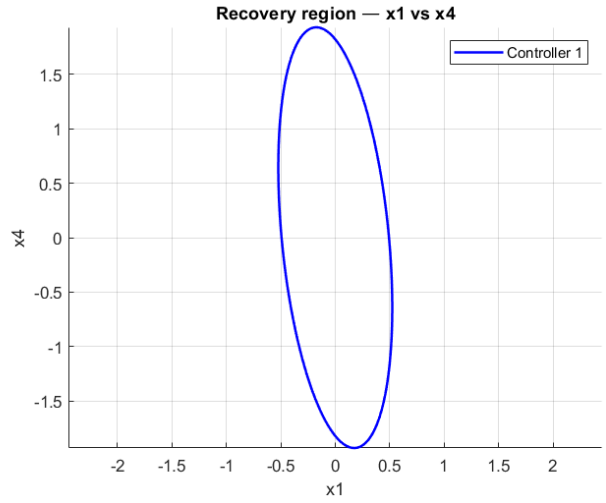


Figure 6: Maximal-volume recovery region on the x_1–x_4 plane (Case 1 and Case 2).

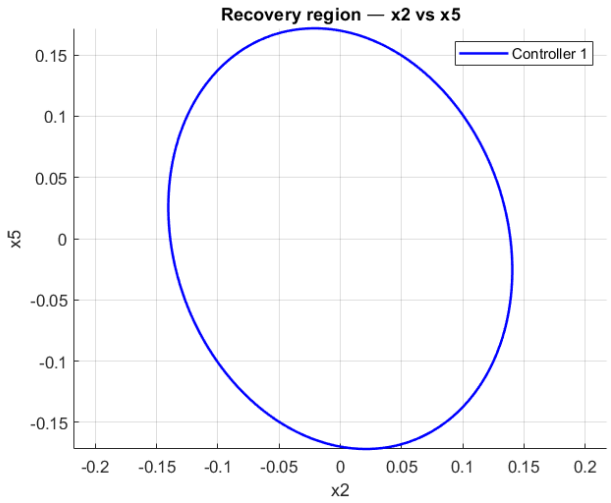


Figure 7: Maximal-volume recovery region on the x_2–x_5 plane (Case 1 and Case 2).

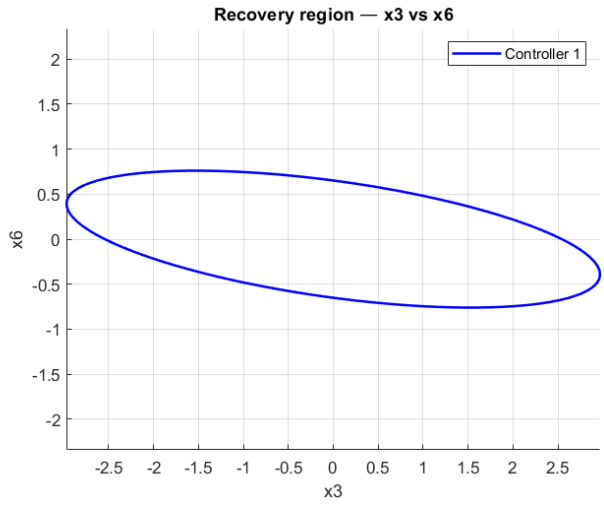


Figure 8: Maximal-volume recovery region on the x_3–x_6 plane (Case 1 and Case 2).

5 Exercise 4: Closed-Loop Simulation with the Exercise 2.3 Controller

In this step, the controller obtained in Exercise 2.3 ($K_{\max_vol}$) is used to simulate the closed-loop helicopter model with initial condition

$$X_0 = [0.479 \;\; 0.0872 \;\; -0.8726 \;\; 0 \;\; 0 \;\; 0]^T, \tag{23}$$

and quadratic weighting matrix

$$Q = \mathrm{diag}(100, 1, 10, 0, 0, 2). \tag{24}$$

The required performance indices are settling time, overshoot, and maximal deviation for x_1, x_2, and x_3, plus settling time of the quadratic error.

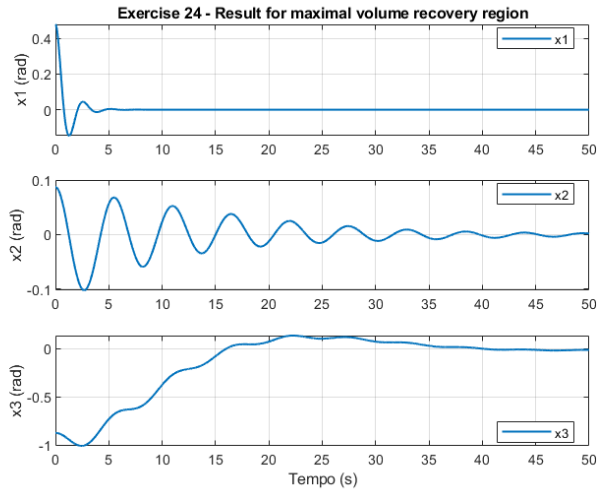


Figure 9: Caption

5.1 Exercise 2.4 Results

Table 4: Simulation results for Exercise 4 using the controller from Exercise 2.3.

Metric	Value
x_1 maximal deviation (rad)	0.47900
x_1 overshoot (rad)	0.14609
x_1 settling time (s)	3.020
x_2 maximal deviation (rad)	0.10391
x_2 overshoot (rad)	0.10391
x_2 settling time (s)	47.570
x_3 maximal deviation (rad)	0.99472
x_3 overshoot (rad)	0.14785
x_3 settling time (s)	34.950
Quadratic error settling time (s)	9.910
Quadratic error tolerance	1.528×10^0

These results indicate fast convergence in x_1, while x_2 and x_3 exhibit significantly longer settling times under the maximal-volume controller.